How Far Can Classical Machine Learning Go on Fashion-MNIST?

A Leak-Free Benchmark of Raw Pixels vs. HOG Features, with Cost and Error Analysis

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***Abstract*—We ask how well classical machine learning classifies the clothing images of Fashion-MNIST, what it costs, and why it fails where it fails. Five classifiers—Logistic Regression, k-Nearest Neighbors, Random Forest, Linear SVM and RBF SVM—plus a majority-class baseline are trained on two representations: raw pixels and Histograms of Oriented Gradients (HOG). Every decision (hyperparameters, HOG configuration, final model) is taken on a fixed, stratified 6,000-image validation split; the 10,000-image test set is touched exactly once per experiment and every access is recorded in an audit log. The model selected on validation, an RBF SVM on HOG features, reaches 91.08% test accuracy (95% CI 90.51–91.62%), significantly better than the same SVM on pixels (McNemar $p < 10^{-5}$). HOG lifts the linear models by 4–5 percentage points (pp) but does not help Random Forest or k-NN, and the most accurate model predicts roughly 2,400× slower than a Random Forest that is only 3.5 pp less accurate. Confusions among four upper-body garments (T-shirt/top, Pullover, Coat, Shirt) account for 56–66% of every model's errors, and Shirt is the hardest class for all ten trained models—exactly as the exploratory analysis predicted before training. These limits of fixed, handcrafted features motivate the convolutional network of Phase 2.**

***Index Terms*—Fashion-MNIST, image classification, HOG, support vector machines, random forest, benchmarking, data leakage, model selection, error analysis**

I. Introduction

An online clothing catalog that tags product photos automatically needs a classifier that is accurate, fast and small. Deep networks are the default answer today, but whether they are *necessary* can only be judged against a strong and honestly evaluated classical baseline. Many published baselines are neither: hyperparameters are tuned against the test set, or models are compared after being tuned on different amounts of data. This report builds such a baseline on Fashion-MNIST [1], a dataset designed as a harder, drop-in replacement for MNIST digits.

We answer four research questions:

- **RQ1**: How accurately can classical ML classify Fashion-MNIST from raw pixels?
- **RQ2**: Do HOG features beat raw pixels, and for which models?
- **RQ3**: Which model offers the best trade-off between accuracy, end-to-end prediction time and model size?
- **RQ4**: Which classes are confused most, and why do handcrafted features fail there?

Contributions. (i) A reproducible benchmark of 12 experiments with a committed validation split and a test-set audit log; (ii) a validation-only selection protocol that also corrects an unfair comparison between models tuned on different amounts of data; (iii) a cost analysis that counts feature-extraction time, not just model time; and (iv) an error analysis, including a paired significance test and an agreement analysis between the two best models, that links model mistakes back to predictions made from the data before any model was trained.

II. Background

Classifiers. *Logistic Regression* is a linear model trained with cross-entropy and an L2 penalty whose strength is set by C (smaller C, stronger regularization). *k-NN* [2] stores the training set and labels an image by its k nearest neighbors. A *Random Forest* [3] averages many decision trees, each grown on a bootstrap sample with a random feature subset at every split. *Support vector machines* [4] find a maximum-margin boundary; the linear variant uses LIBLINEAR [5] and the kernel variant uses the RBF kernel

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma \left\|\mathbf{x} - \mathbf{x}'\right\|^2\right) \tag{1}$$

to draw non-linear boundaries. Its prediction cost grows with the number of support vectors, since every test point is compared with each of them.

HOG. Histograms of Oriented Gradients [6] describe an image by the directions of its edges. The image is divided into cells of $c \times c$ pixels; each cell receives a histogram of o gradient-orientation bins weighted by gradient magnitude, and overlapping blocks of $b \times b$ cells are normalized together, which makes the descriptor robust to brightness and contrast. For a 28×28 image the descriptor length is

$$d = \left(\tfrac{28}{c} - b + 1\right)^2 b^2\, o. \tag{2}$$

Leakage and selection bias. Using test data for any decision makes the reported test score optimistic [7]. Even with a clean test set, picking the best of many configurations on validation makes the *winning validation score* optimistic [8]; we therefore report the untouched test score as the headline number.

Software. All models use scikit-learn [9]; HOG uses scikit-image [10].

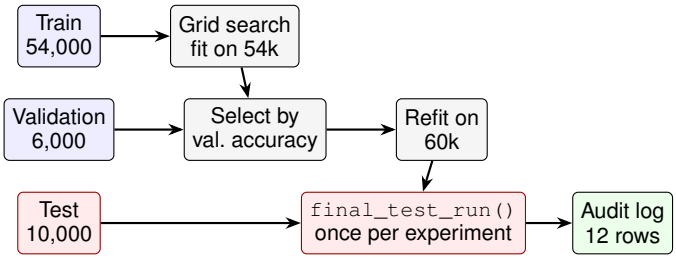


Fig. 1. Evaluation protocol. Every decision is taken on validation; the test set (red) is read by a single function, once per experiment, and each read is logged.

III. Data and Experimental Protocol

A. Dataset and Integrity Checks

Fashion-MNIST contains 70,000 grayscale 28×28 images in 10 balanced classes—T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag and Ankle boot—split into 60,000 training and 10,000 test images. The four official files are downloaded over HTTPS and verified against the published MD5 checksums. Automated checks confirm the array shapes, labels in $\{0, \ldots, 9\}$, no missing values, exactly 6,000 training and 1,000 test images per class, and no exact duplicate images within or across the training and test sets.

B. Fixed Split

A stratified validation set of 6,000 images (600 per class, seed 42) is carved from the official training set, leaving 54,000 training images. The validation indices are committed to the repository (MD5 `54ff6b78b27109cdfc3ebe7204040a07`), so every team member uses an identical split. The official test set is the final, untouched evaluation set (Fig. 1).

C. Leakage Controls

1) The test set is read by exactly one function, `final_test_run()`, once per experiment. Each call appends a row to `test_log.csv`; the final log has exactly 12 rows, one per experiment.
2) Hyperparameters, HOG settings and the final model are chosen on validation accuracy only.
3) Feature scaling (`StandardScaler`) lives inside each scikit-learn `Pipeline`, so it is fitted only on the rows a model is trained on, including within cross-validation folds.
4) Exploratory analysis uses the 54,000-image training split only.
5) Error analysis and significance tests read the predictions saved at test time; no model predicts on the test set again.

D. Procedure per Experiment

For each model–feature pair we (1) fit every grid combination on the 54k training split and score it on validation; (2) take the combination with the highest validation accuracy; (3) refit it on all 60,000 training and validation images; and (4) evaluate it once on test, recording accuracy, macro-F1, per-class precision and recall, the confusion matrix, training time, model-only prediction time per 1,000 images (best of three runs), and model size (bytes of the pickled model). Test accuracies are reported with Wilson 95% confidence intervals; with $n = 10{,}000$ these are about ± 0.6 pp wide for every trained model.

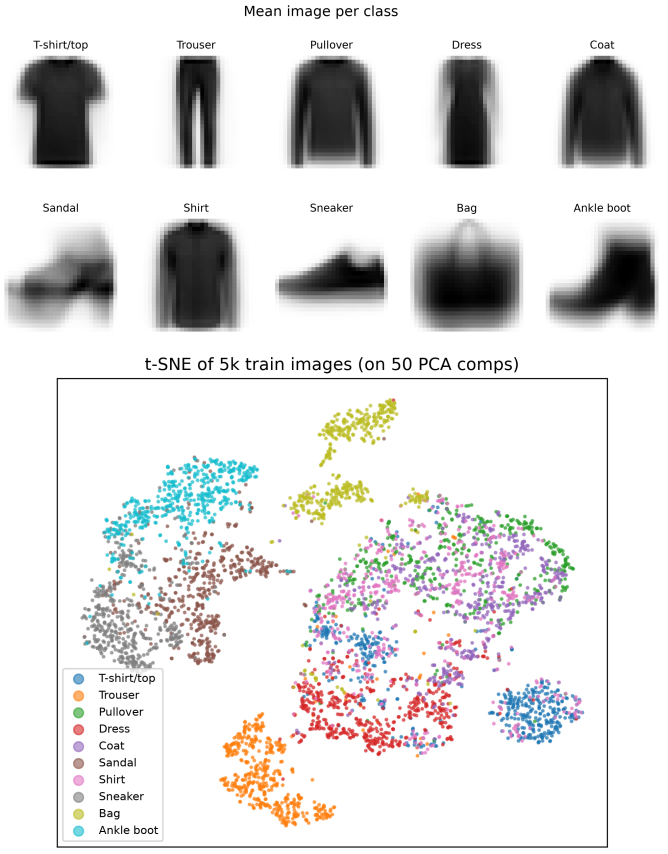


Fig. 2. Top: mean image of each class (training split). Bottom: t-SNE embedding of a training subsample. Pullover, Coat and Shirt overlap; Trouser and Bag are well separated.

IV. Exploratory Data Analysis

Pixel values are dominated by background: PCA needs 84 of 784 components to retain 90% of the variance and 188 to retain 95%, so raw pixels are highly redundant. Fig. 2 shows the class-mean images and a t-SNE [11] embedding of the training split. The mean images of the upper-body garments are nearly identical: cosine similarity is 0.987 for Coat–Shirt, 0.984 for Pullover–Coat, 0.983 for Pullover–Shirt and 0.950 for T-shirt/top–Shirt. In the t-SNE map, Pullover, Coat and Shirt form one mixed cloud, while Trouser and Bag form clean, isolated clusters.

Predictions made before training. Before any model was trained, the EDA led us to predict that (a) Shirt would have the lowest recall and (b) most errors would fall within the upper-body group. Section VIII tests both.

TABLE I
HOG SWEEP: PROXY VALIDATION ACCURACY
(7 OF 12 CONFIGURATIONS SHOWN; SELECTED SETTING IN BOLD)

Orient. o	Cell c	Block b	Dim. d	Val. acc.
6	4	2	864	0.8757
9	4	2	1,296	0.8783
12	**4**	**2**	**1,728**	**0.8818**
12	4	1	588	0.8630
12	7	2	432	0.8357
12	7	1	192	0.8010
6	7	1	96	0.7612

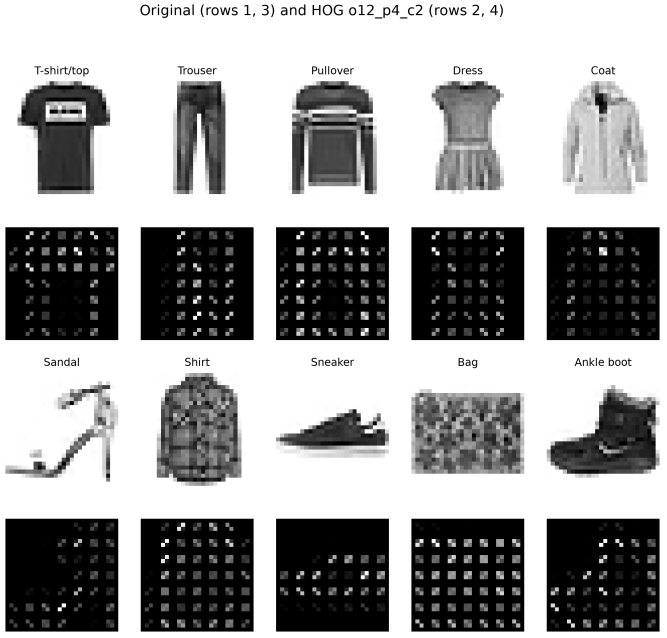


Fig. 3. One training image per class beside its HOG descriptor. The four tops yield similar torso-and-sleeve edge patterns; footwear and trousers have distinctive edge layouts.

V. FEATURE REPRESENTATIONS

Pixels. Each image is flattened to a 784-dimensional vector in $[0, 1]$. Flattening discards the 2-D neighborhood: a garment shifted by two pixels becomes a very different vector.

HOG configuration sweep. We evaluated $3 \times 2 \times 2 = 12$ HOG configurations: orientations $o \in \{6, 9, 12\}$, cell size $c \in \{4, 7\}$ pixels and block size $b \in \{1, 2\}$ cells. Each was scored with a fast proxy—a linear SVM trained by SGD with hinge loss—fitted on training features and scored on validation (Table I). Small cells win by 4–8 pp, block normalization adds 2–3 pp, and more orientation bins help slightly. The selection rule, fixed in advance, was *highest validation accuracy; within 0.3 pp prefer the shorter vector*. The winner ($o = 12$, $c = 4$, $b = 2$, $d = 1{,}728$ by (2); 0.8818) beats the 1,296-dimensional runner-up (0.8783) by 0.35 pp, so it was kept. Extraction costs about 0.22 s per 1,000 images.

Scaling. Logistic Regression, the SVMs and k-NN are sensitive to feature scale, so they standardize inside their pipeline. A Random Forest splits on one feature at a time by threshold, which is invariant to monotone rescaling, so it uses no scaler.

TABLE II
VALIDATION GRIDS AND SELECTED SETTINGS

Model	Grid	Selected (pixels / HOG)
Log. Reg.	$C \in \{0.01, 0.1, 1, 10\}$	$C = 0.01$ / $C = 0.01$
k-NN	$k \in \{1, 3, 5, 7, 9\}$; uniform / distance; Euclidean / Manhattan	k=3, distance, Manhattan / k=3, distance, Euclidean
Rand. Forest	trees $\{100, 300\}$; depth $\{\text{None}, 20, 40\}$; features $\{\sqrt{p}, \log_2 p\}$	300, depth 20, $\sqrt{p}$ / 300, full depth, $\sqrt{p}$
Linear SVM	$C \in \{0.01, 0.1, 1\}$	$C = 0.01$ / $C = 0.01$
RBF SVM	$C \in \{0.1, 1, 10\}$; $\gamma \in \{\text{scale}, 0.01, 0.001\}$ (10k subset)	C=10, γ=0.001 / C=10, γ=scale

VI. MODELS AND TUNING

Table II lists the validation grids and the selected settings. Two practical constraints shaped tuning on an 8 GB laptop. First, kernel-SVM training scales super-linearly with the number of samples, so the RBF SVM's 9-point grid was searched on a stratified 10,000-image subset (1,000 per class), and only the selected setting was refit on the full data. Second, Linear SVM fits on 54,000 images took 105–924 s, growing sharply with C.

Making the validation comparison fair. Because the RBF SVM was tuned on 10,000 images, its tuning-time validation scores (0.8793 pixels, 0.8867 HOG) came from far less training data than every other model's (54,000) and understated it. *Before* choosing the final model, we fixed a correction: refit the *already selected* RBF settings on the full 54k training split and score them on the same validation set—no new search and no test data. This gave 0.9082 (pixels) and 0.9157 (HOG). On equal footing, the validation ranking is RBF SVM / HOG (0.9157) $>$ RBF SVM / pixels (0.9082) $>$ Logistic Regression / HOG (0.8927) $>$ Linear SVM / HOG (0.8900).

Observations. The best C lies on the grid boundary for Logistic Regression and Linear SVM (0.01, the smallest) and for the RBF SVM (10, the largest); we did not widen the grids afterwards (Section IX). The RBF bandwidth behaves as theory predicts. After standardization, scikit-learn's `scale` setting equals $1/d$: about 0.0013 for pixels and 0.00058 for HOG. The selected values sit at this scale, whereas $\gamma = 0.01$ is 17$\times$ too narrow for HOG; the kernel in (1) then treats almost every pair of images as dissimilar, and HOG validation accuracy collapses to 11–29%.

VII. RESULTS

Table III reports all 12 experiments. Pixel-feature results are in line with the scikit-learn baselines published with the dataset [1].

TABLE III
RESULTS OF ALL 12 EXPERIMENTS

Model	Features	Val. acc.	Test acc.	95% CI (%)	Macro-F1	Train (s)	Inference (ms/1k)	Size
Dummy (majority)	Pixels	0.1000	0.1000	9.4–10.6	0.0182	0.00	0.00[a]	709 B
	HOG	0.1000	0.1000	9.4–10.6	0.0182	0.16	0.01[a]	709 B
Logistic Regression	Pixels	0.8642	0.8472	84.0–85.4	0.8464	n/a[b]	n/a[b]	51.4 KB
	HOG	0.8927	0.8886	88.2–89.5	0.8885	n/a[b]	n/a[b]	111.8 KB
k-NN	Pixels	0.8747	0.8611	85.4–86.8	0.8614	0.49	6,926.61	188.7 MB
	HOG	0.8728	0.8631	85.6–87.0	0.8623	1.29	731.13	415.2 MB
Random Forest	Pixels	0.8867	0.8756	86.9–88.2	0.8741	32.83	12.16	366.5 MB
	HOG	0.8735	0.8703	86.4–87.7	0.8692	68.53	19.02	500.7 MB
Linear SVM	Pixels	0.8583	0.8378	83.0–84.5	0.8353	99.24	1.22	82.7 KB
	HOG	0.8900	0.8898	88.4–89.6	0.8892	162.04	2.62	180.8 KB
RBF SVM	Pixels	0.9082	0.8970	89.1–90.3	0.8967	195.67	4,766.94	125.1 MB
	HOG	**0.9157**	**0.9108**	**90.5–91.6**	**0.9108**	451.43	28,876.96[c]	366.2 MB

Val. acc.: model trained on 54k (RBF SVM refit on 54k for comparability). Test: final model trained on 60k, evaluated once; CI is the Wilson 95% interval. Inference: model-only time per 1,000 test images; HOG extraction adds ≈216 ms. Best result in bold.
[a] Measured on a different machine; excluded from speed comparisons.
[b] Not comparable (runs overlapped other workloads); accuracy unaffected.
[c] 26,343 support vectors. About 2× slower than expected from the pixel model, so treated as an upper bound.
Timing hardware: k-NN on an Apple M3; Random Forest and SVMs on an Apple M3 Pro.

TABLE IV
END-TO-END DEPLOYMENT TRADE-OFF
(TIME INCLUDES HOG EXTRACTION WHERE USED)

Model / features	Test acc.	ms per 1k	Size
Random Forest / pixels	87.56%	≈12	366.5 MB
Linear SVM / HOG	88.98%	≈219	180.8 KB
RBF SVM / pixels	89.70%	≈4,767	125.1 MB
RBF SVM / HOG	91.08%	≤29,093	366.2 MB

A. RQ1: Raw Pixels

The best pixel model is the RBF SVM at 89.70%, followed by Random Forest (87.56%) and k-NN (86.11%); the linear models reach only 83.78–84.72%. A single pixel records brightness at one location, so a linear boundary in pixel space cannot express shape, and models that build non-linear combinations of pixels do markedly better.

B. RQ2: HOG vs. Pixels

HOG helps the linear models most: $+5.20$ pp for Linear SVM ($83.78 \rightarrow 88.98\%$) and $+4.14$ pp for Logistic Regression ($84.72 \rightarrow 88.86\%$), turning both into competitive models. It lifts the RBF SVM by $+1.38$ pp to 91.08%. It does not help Random Forest (-0.53 pp) or k-NN ($+0.20$ pp, well inside the CI) (Fig. 4, left). A plausible explanation for the forest is that each split inspects a single feature, and one raw pixel is more informative on its own than one HOG bin, which describes a single edge direction in one small block; we did not test this hypothesis directly.

C. RQ3: Accuracy, Speed and Size

A fair speed comparison must include feature extraction (Table IV). Three options stand out:

- **Lowest latency above 87%:** Random Forest on pixels—87.56% in about 12 ms per 1,000 images, but a 366 MB model.
- **Best accuracy per byte:** Linear SVM on HOG—88.98% from a 181 KB model, about 2,000× smaller than the forest; its 219 ms per 1,000 images is almost entirely HOG extraction.
- **Highest accuracy:** RBF SVM on HOG—91.08%, but up to 29 s per 1,000 images, roughly 2,400× slower than the forest for $+3.5$ pp, because each prediction is compared with all 26,343 support vectors.

k-NN is both slower and less accurate than the Random Forest, so it is dominated and never the right choice here. For the catalog use case, the Linear SVM on HOG is the most practical classical model; the RBF SVM sets the accuracy bar for Phase 2.

D. Selected Model, Significance and Stability

The model selected on validation is the RBF SVM on HOG (0.9157); the test score played no part in this choice. Its validation lead over the pixel RBF SVM (0.75 pp) is only about twice the standard error of a 6,000-image validation estimate (≈0.36 pp), so on validation alone it is suggestive rather than conclusive. The held-out test set confirms it: on the paired test predictions, the HOG model is wrong on 377 images the pixel model gets right, and right on 515 the pixel model gets wrong. A two-sided McNemar test [12] with continuity correction gives $\chi^2 = 21.0$, $p = 4.5 \times 10^{-6}$ (exact binomial $p = 4.3 \times 10^{-6}$), so the 1.38 pp gain is statistically significant.

A stratified 5-fold cross-validation on a 10,000-image subset of the training and validation data gives 0.8775 ± 0.0054 (folds 0.8765, 0.8860, 0.8715, 0.8725, 0.8810). The small spread indicates the selected configuration is stable across partitions;

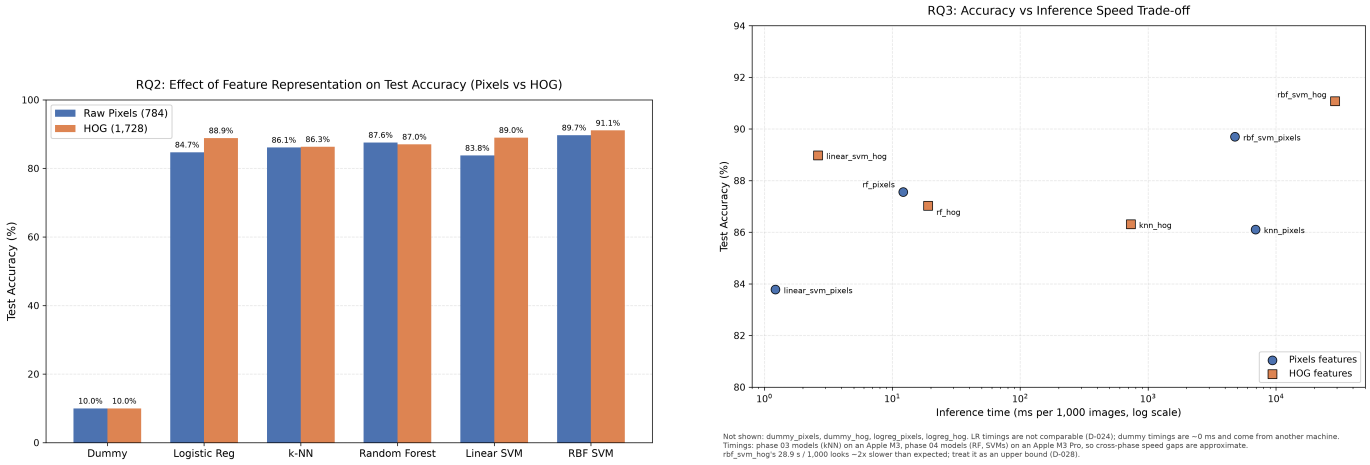


Fig. 4. Left: test accuracy with pixel vs. HOG features (RQ2). Right: test accuracy vs. model-only prediction time per 1,000 images on a log scale (RQ3). Logistic Regression and the dummy baseline are excluded from the speed plot.

the lower mean is expected because each fold trains on only 8,000 images.

VIII. Error Analysis (RQ4)

Per-class recall. Shirt is the hardest class for *all ten* trained models, with recall from 50.9% (Linear SVM, pixels) to 75.2% (RBF SVM, HOG), followed by Pullover (73.5–84.9%) and Coat (76.4–86.3%). Trouser, Bag, Sandal, Sneaker and Ankle boot are easy for every model (Fig. 5, left). This confirms prediction (a) from Section IV.

Dominant confusions. From the selected model's confusion matrix (1,000 test images per class), the five largest off-diagonal entries are Shirt → T-shirt/top (104), T-shirt/top → Shirt (91), Pullover → Shirt (66), Pullover → Coat (61) and Shirt → Coat (59) (Fig. 5, right). Across all ten trained models, confusions within {T-shirt/top, Pullover, Coat, Shirt} make up 56–66% of all errors, confirming prediction (b).

Inspecting the mistakes. Fig. 6 shows the first 16 test shirts, in test-set order and not hand-picked, that the selected model labels T-shirt/top. Many are short-sleeved, collarless or patterned. At 28×28 the collar and buttons that define a shirt cover only a handful of pixels, and several images are ambiguous even to a human.

Do the two best models fail on the same images? Table V cross-tabulates the saved predictions of the RBF SVM on HOG and on pixels. HOG fixes 515 images and breaks 377—a net gain of 138, exactly the 1.38 pp test gap. Of the 515 images both models miss, 391 (76%) receive the *same* wrong label and 397 (77%) are upper-body garments. When two models built on very different representations make the same mistake, the image itself is the likely cause, not the model. This residual error is what a stronger representation must address.

IX. Discussion and Limitations

- **Grid boundaries.** Three models selected a value on the edge of their grid, so wider grids might add a little accuracy. We kept the planned grids rather than spend additional test evaluations.
- **Optimistic validation.** Validation accuracy exceeds test accuracy for all ten trained models, by 0.02–2.05 pp, as expected when the best of several settings is selected on validation [8]. The model ranking is unchanged, and we report the untouched test score.
- **RBF tuning on a subset.** The RBF grid was searched on 10k images; settings that are best at 10k may not be best at 54k, so the RBF results are, if anything, a lower bound.
- **Timing.** Timings come from two laptops (Apple M3 and M3 Pro) and two Logistic Regression timings are excluded; only large gaps ($\geq 10\times$) should be read as firm.
- **Cross-validation** used a 10k subset for time reasons, so it measures stability, not final accuracy.
- **Single seed.** Random Forest results come from one seed; seed variance was not measured.

TABLE V
Agreement of the Two Best Models on the 10,000 Test Images

	RBF SVM / pixels		
RBF SVM / HOG	correct	wrong	Total
correct	8,593	515	9,108
wrong	377	515	892
Total	8,970	1,030	10,000

X. Conclusion and Future Work

A carefully tuned classical pipeline reaches 91.08% test accuracy on Fashion-MNIST with an RBF SVM on HOG features, selected without using the test set for any decision, and significantly better than the same model on raw pixels. HOG is the single largest improvement for linear models but does nothing for trees or nearest neighbors, and the most

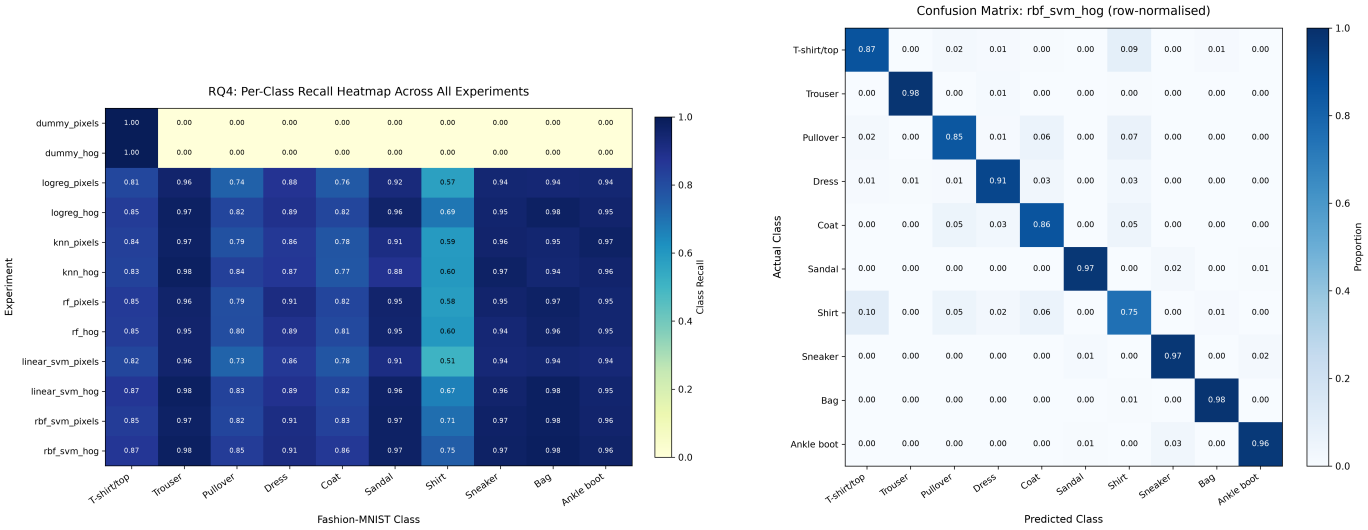


Fig. 5. Left: per-class recall of all 12 experiments. Right: row-normalized confusion matrix of the selected model (RBF SVM, HOG). The upper-body garments form the dominant block of errors.



Fig. 6. The first 16 (of 104) test images of class Shirt predicted as T-shirt/top by the selected model.

accurate model is also by far the slowest; a 181 KB Linear SVM on HOG is the most practical deployment choice. The remaining errors concentrate in four upper-body garments whose distinguishing details—collars, buttons, cuffs—occupy only a few pixels and are poorly captured by fixed edge histograms. Both outcomes were predicted from the data before training.

Phase 2. We will train a small convolutional network [13] (2–3 convolutional blocks with pooling, dropout and early stopping) on the *same* split and score it with the *same* evaluation code, so it drops directly into Table III and can be compared to the RBF SVM with the same McNemar test. Our hypothesis is that learned local filters capture these small details and exceed 91.08%, chiefly by reducing confusions within {Shirt, T-shirt/top, Pullover, Coat}.

Reproducibility. Code, the committed split, all result files, figures and a decision log recording the reason for every choice are available at https://github.com/rajdeep-2004/fashion-image-classification-benchmark. A fresh clone regenerates every reported result from the saved artifacts in under two minutes.

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